

TRIV: Design and Construction Firm

**APPLICATIONS DEVELOPMENT AND EMERGING TECHNOLOGIES 2
(ENTERPRISE-BACK-END)**

ICS2609

2ITB

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PHP CODES

CRUD IMPLEMENTATION (For Projects)

```
<?php
require_once '../classes/Auth.php';
require_once '../classes/Database.php';

Auth::requireLogin();
Auth::requireRole('admin');

$db = new Database();
$conn = $db->connect();
$project = new Project($conn);

// Handle form submissions
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    if (isset($_POST['action'])) {
        switch ($_POST['action']) {
            case 'create':
                $imageName = null;
                if (isset($_FILES['image']) && $_FILES['image']['error'] === 0) {
                    $imageName = handleImageUpload($_FILES['image']);
                }
                $project->create($_POST['title'], $_POST['description'], $_POST['location'],
$_POST['category'], $imageName);
                break;

            case 'update':
                $imageName = null;
                if (isset($_FILES['image']) && $_FILES['image']['error'] === 0) {
                    $imageName = handleImageUpload($_FILES['image']);
                }
                $project->update($_POST['id'], $_POST['title'], $_POST['description'],
$_POST['location'], $_POST['category'], $imageName);
                break;

            case 'delete':
                $project->delete($_POST['id']);
                break;
        }
    }
}
```

```

    }
}

$projects = $project->getAll();

// Image upload handler
function handleImageUpload($file) {
    // ...continued
}
?>

```

OOP (Database Connection Logic with Encapsulation)

```

<?php
class Database {
    private $host = "localhost";
    private $db_name = "triv_db";
    private $username = "root";
    private $password = "";
    private $conn;

    public function connect() {
        try {
            $this->conn = new PDO(
                "mysql:host=$this->host;dbname=$this->db_name",
                $this->username,
                $this->password
            );
            $this->conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
            return $this->conn;
        } catch (PDOException $e) {
            die("Database connection failed: " . $e->getMessage());
        }
    }
}

class Project {
    private $conn;
    private $table = 'projects';
}

```

```

    public function __construct($db) {
        $this->conn = $db;
    }

    public function getAll() {
        $sql = "SELECT * FROM " . $this->table . " ORDER BY created_at DESC";
        $stmt = $this->conn->prepare($sql);
        $stmt->execute();
        return $stmt->fetchAll(PDO::FETCH_ASSOC);
    }

    public function getByCategory($category) {
        $sql = "SELECT * FROM " . $this->table . " WHERE category = :category ORDER
BY created_at DESC";
        $stmt = $this->conn->prepare($sql);
        $stmt->bindParam(':category', $category);
        $stmt->execute();
        return $stmt->fetchAll(PDO::FETCH_ASSOC);
    }

    // ..more

```

OOP (Session and Cookies Logic with Abstraction)

```

<?php
class Auth {
    private static $sessionStarted = false;

    public static function startSession() {
        if (!self::$sessionStarted && session_status() === PHP_SESSION_NONE) {
            ini_set('session.cookie_httponly', 1);
            ini_set('session.cookie_secure', isset($_SERVER['HTTPS']));
            ini_set('session.use_strict_mode', 1);
            session_start();
            self::$sessionStarted = true;
        }
    }
}

```



```

        if (!isset($_SESSION['last_regeneration']) || time() - $_SESSION['last_regeneration'] >
300) {

            self::regenerateSession();

        }

    } // ...more

private static function regenerateSession() {
    session_regenerate_id(true);
    $_SESSION['last_regeneration'] = time();
} // ...more

public static function login($userData, $remember = false) {
    self::startSession();
    $_SESSION['user_id'] = $userData['id'];
    $_SESSION['user_name'] = $userData['name'];
    $_SESSION['user_email'] = $userData['email'];
    $_SESSION['user_role'] = $userData['role'];
    $_SESSION['login_time'] = $_SESSION['last_activity'] = time();

    if ($remember) {
        $exp = time() + (86400 * 30);
        setcookie('remember_user_id', $userData['id'], $exp, "/", "", isset($_SERVER['HTTPS']),
true);

        setcookie('remember_user_token', self::generateRememberToken($userData['id']), $exp,
"/", "", isset($_SERVER['HTTPS']), true);
    }

    self::regenerateSession();
} // ...more

public static function logout() {
    self::startSession();
    setcookie('remember_user_id', '', time() - 3600, "/");
    setcookie('remember_user_token', '', time() - 3600, "/");
    $_SESSION = [];

    if (ini_get("session.use_cookies")) {
        $p = session_get_cookie_params();
        setcookie(session_name(), '', time() - 42000, $p["path"], $p["domain"], $p["secure"],

```

```

$p["httponly"]);
    }

    session_destroy();
    header("Location: ../public/login.php");
    exit();
} // ...more

public static function isLoggedIn() {
    self::startSession();

    if (isset($_SESSION['user_id'], $_SESSION['last_activity'])) {
        if (time() - $_SESSION['last_activity'] > 1800) {
            self::logout(); return false;
        }
        $_SESSION['last_activity'] = time();
        return true;
    }

    if (isset($_COOKIE['remember_user_id'], $_COOKIE['remember_user_token'])) {
        return self::validateRememberToken($_COOKIE['remember_user_id'],
$_COOKIE['remember_user_token']);
    }

    return false;
} // ...more

public static function requireLogin() {
    if (!self::isLoggedIn()) {
        header("Location: ../public/login.php?redirect=" . urlencode($_SERVER['REQUEST_URI']));
        exit();
    }
} // ...more

public static function requireAdmin() {
    self::requireLogin();
    if (!self::isAdmin()) {
        header("Location: ../public/index.php?error=access_denied");
        exit();
    }
} // ...more

```

```

public static function requireRole($requiredRole) {
    self::requireLogin();
    $role = self::getUserRole();
    if ($role !== $requiredRole && $role !== 'admin') {
        header("Location: ../public/index.php?error=insufficient_permissions");
        exit();
    }
} // ...more

public static function isAdmin() {
    return self::getUserRole() === 'admin';
} // ...more

public static function getUserId() {
    self::startSession();
    return $_SESSION['user_id'] ?? null;
} // ...more

public static function getUsername() {
    self::startSession();
    return $_SESSION['user_name'] ?? null;
} // ...more

public static function getUserEmail() {
    self::startSession();
    return $_SESSION['user_email'] ?? null;
} // ...more

public static function getUserRole() {
    self::startSession();
    return $_SESSION['user_role'] ?? null;
} // ...more

public static function redirectBasedOnRole($role = null) {
    $role = $role ?? self::getUserRole();
    header("Location: " . ($role === 'admin' ? "../admin/dashboard.php" :
"../public/index.php"));
    exit();
} // ...more

```

```

private static function generateRememberToken($userId) {
    return hash('sha256', $userId . time() . random_bytes(16));
} // ...more

private static function validateRememberToken($userId, $token) {
    if (!empty($userId) && !empty($token)) {
        try {
            $user = (new User((new Database())->connect()))->getUserById($userId);
            if ($user) {
                $_SESSION['user_id'] = $user['id'];
                $_SESSION['user_name'] = $user['name'];
                $_SESSION['user_email'] = $user['email'];
                $_SESSION['user_role'] = $user['role'];
                $_SESSION['login_time'] = $_SESSION['last_activity'] = time();
                return true;
            }
        } catch (Exception $e) {
            error_log("Token error: " . $e->getMessage());
        }
    }
    return false;
} // ...more

public static function checkAdminAccess() {
    self::requireAdmin();
} // ...more

public static function getLoginRedirectUrl() {
    return $_GET['redirect'] ?? null;
} // ...more
}
?>

```

DASHBOARD (Session and CRUD Implementation)

```

<?php
require_once '../classes/Auth.php';
require_once '../classes/Database.php';

// Ensure only admins can access this page

```

```

Auth::checkAdminAccess();

$db = new Database();
$conn = $db->connect();

$project = new Project($conn);
$user = new User($conn);
$service = new Service($conn);
$contactInquiry = new ContactInquiry($conn);
$job = new Job($conn);
$jobApplication = new JobApplication($conn);

// Get dashboard statistics
$projectCount = $project->getCount();
$userCount = $user->getCount();
$serviceCount = $service->getCount();
$inquiryCount = $contactInquiry->getCount();
$pendingInquiries = $contactInquiry->getPendingCount();
$recentInquiries = $contactInquiry->getRecentInquiries();

$jobCount = $job->getCount();
$applicationCount = $jobApplication->getCount();
$pendingApplications = $jobApplication->getPendingCount();
$recentApplications = $jobApplication->getRecentApplications(3);

$username = Auth::getUserName();
?>

```

LOG IN AND REGISTER (Session and Cookies Implementation PHP Forms)

```

<?php
require_once '../classes/Database.php';
require_once '../classes/Auth.php';

Auth::startSession();

```

```

// Redirect if already logged in
if (Auth::isLoggedIn()) {
    Auth::redirectBasedOnRole();
}

$db = new Database();
$conn = $db->connect();
$user = new User($conn);

$error = "";
$success = "";

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    $remember = isset($_POST['remember']);

    if (empty($email) || empty($password)) {
        $error = "Please fill in all fields.";
    } else {
        $loginResult = $user->login($email, $password);

        if ($loginResult['success']) {
            Auth::login($loginResult['user'], $remember);

            // Check for redirect URL
            $redirectUrl = Auth::getLoginRedirectUrl();
            if ($redirectUrl) {
                header("Location: " . $redirectUrl);
            } else {
                Auth::redirectBasedOnRole($loginResult['user']['role']);
            }
            exit();
        } else {
            $error = $loginResult['message'];
        }
    }
}
}

```



```
        <button type="submit" class="login-btn">Login</button>
    </form>
```

INCORPORATION OF DATABASE (../public/projects.php)

```
<?php include 'functions.php'; ?>
<?php
require_once '../classes/Database.php';

$db = new Database();
$conn = $db->connect();

$project = new Project($conn);

$constructionProjects = $project->getByCategory('construction');
$architecturalProjects = $project->getByCategory('architectural-design');
$renovationProjects = $project->getByCategory('renovation');
$interiorProjects = $project->getByCategory('interior-design');
?>

//...

<!-- Renovation Section -->
<div id="renovation-section" class="content-section-projects
hidden-projects">
    <h2 class="content-title-projects">Renovation</h2>
    <p class="content-text-projects">TRANSFORMING SPACES.</p>
    <p class="content-text-projects">Our renovation services breathe new
life into existing structures, enhancing both form and function.</p>

    <h3 class="content-subtitle-projects"></h3>
    <p class="content-text-projects">Our renovation approach begins with a
thorough assessment of the existing structure, identifying both challenges and
opportunities. We then develop a comprehensive plan that respects the original
character of the building while incorporating modern improvements and design
elements.</p>
```



```

        <h3 class="content-subtitle-projects"></h3>
        <p class="content-text-projects">Our renovation services include
structural upgrades, space reconfiguration, facade improvements, and systems
modernization. We specialize in both residential and commercial renovations, with
particular expertise in historic building restoration and adaptive reuse
projects.</p>

        <h3 class="content-subtitle-projects"></h3>
        <p class="content-text-projects"></p>

        <div class="projects-grid">
<?php foreach ($renovationProjects as $proj): ?>
    <div class="project-card">
        <div class="project-image" style="background-image: url(../assets/images/<?=
htmlspecialchars($proj['image']) ?>);"></div>
        <div class="project-details">
            <h3 class="project-name"><?= htmlspecialchars($proj['title']) ?></h3>
            <p class="project-position"><?= htmlspecialchars($proj['location'])
?></p>
            <p class="project-bio"><?= htmlspecialchars($proj['description']) ?></p>
        </div>
    </div>
<?php endforeach; ?>

        </div>
    </div>

```

PHP FORMS (../public/contact.php)

```

<section class="contact-main">
    <div class="contact-container">
        <div class="contact-form-section">
            <form action="contact.php" method="post"
enctype="multipart/form-data">
                <div class="form-group">

```

```

        <input type="text" name="name" placeholder="Name:" required>
    </div>
    <div class="form-group">
        <input type="email" name="email" placeholder="Email:"
required>
    </div>
    <div class="form-group">
        <input type="tel" name="mobile" placeholder="Mobile:"
required>
    </div>
    <div class="form-group">
        <textarea name="message" placeholder="Your Message" rows="6"
required></textarea>
    </div>
    <div class="form-group file-upload">
        <label for="plan-upload" class="file-label">
            <span>Drop Plan Here</span>
            <input type="file" id="plan-upload" name="plan"
class="file-input">
        </label>
    </div>
    <button type="submit" class="submit-btn">Submit</button>
</form>
</div>

<div class="contact-info-section">
    <h2>TRIV Design & Construction</h2>

    <div class="contact-detail">
        <div class="contact-icon">
            
        </div>
        <p>322 National Highway, Masaya Rosario, Batangas</p>
    </div>

    <div class="contact-detail">
        <div class="contact-icon">
            
        </div>
    </div>

```

```
<p>09087420857</p>  
</div>  
</div>  
</div>  
</section>
```

ERD DIAGRAM

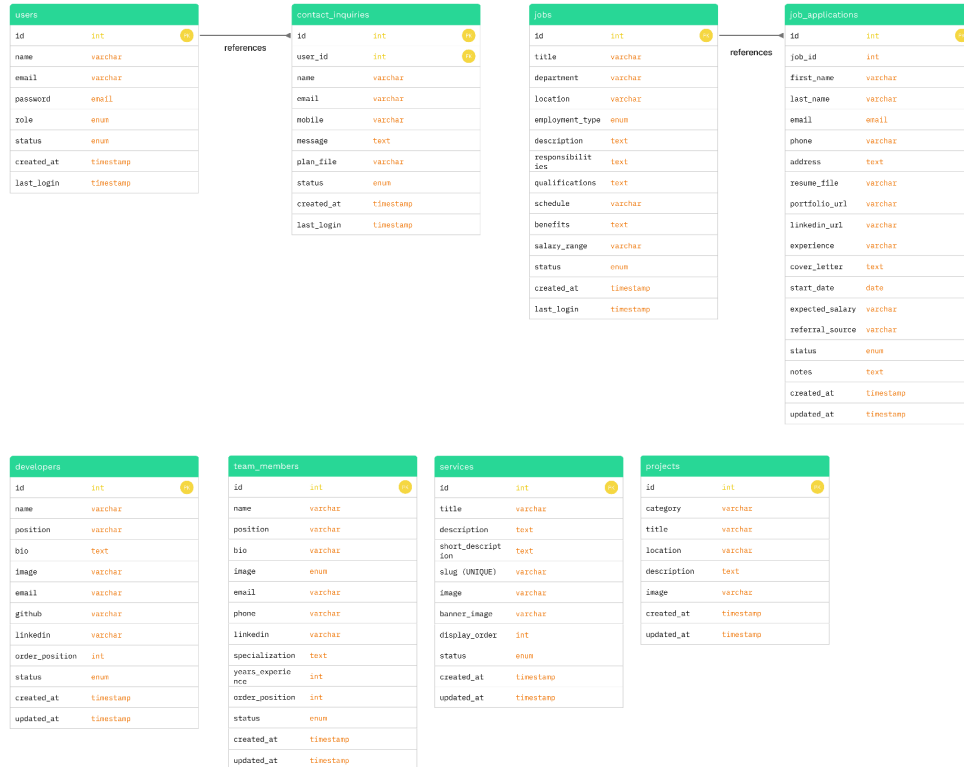


Figure 1.1 - ERD Diagram

The core of the database architecture is centered around the **users** table, which securely stores user credentials and essential metadata, including the user's full name, email address, password, role designation (e.g., *admin* or *client*), and login status. This table maintains a **one-to-many relationship** with the **contact_inquiries** table, allowing each user to submit multiple inquiries. Each record in the **contact_inquiries** table captures detailed information such as the sender's contact details, a personalized message, and, optionally, an uploaded plan file—particularly valuable for architecture or construction-related discussions.

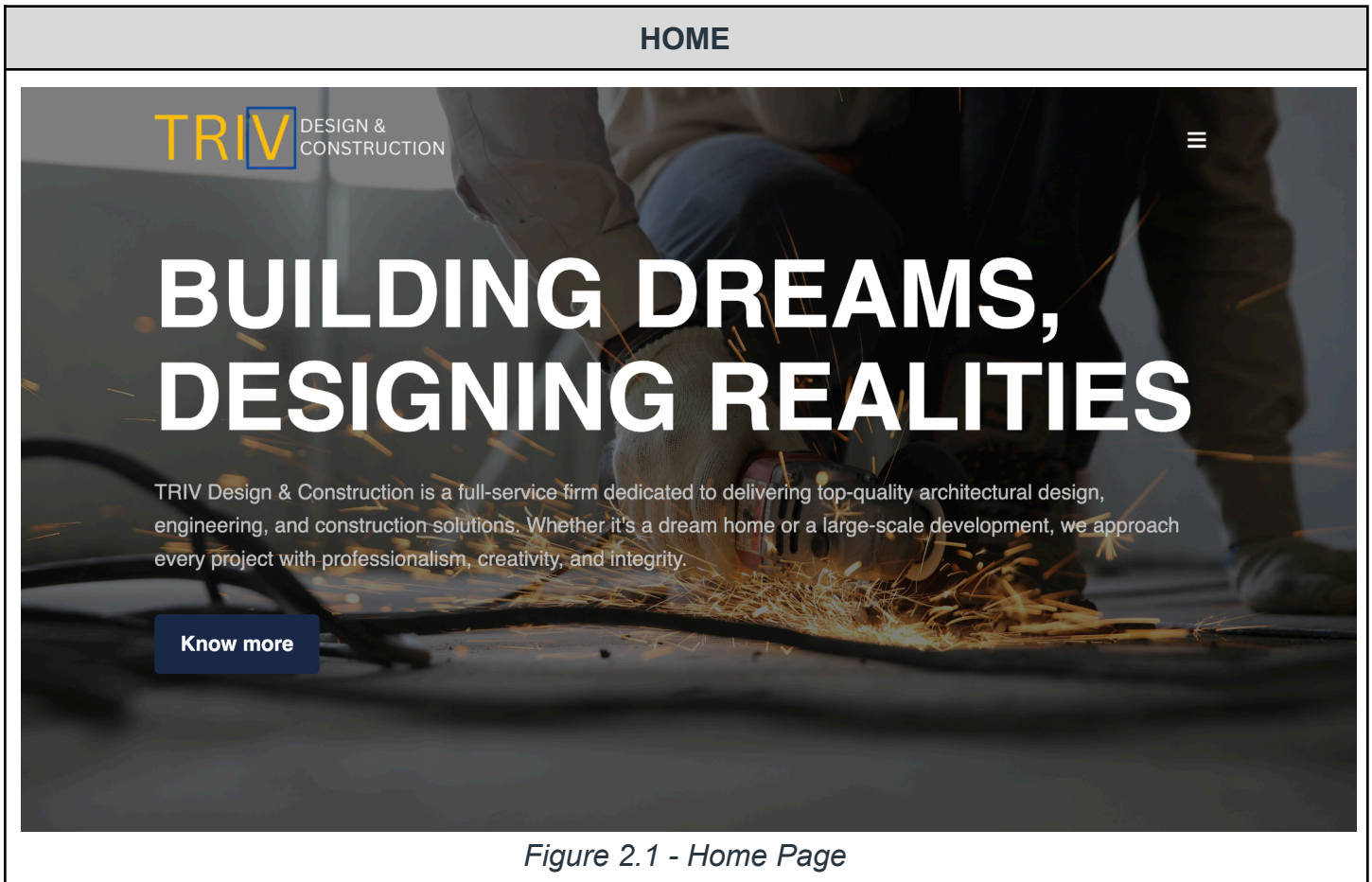
In parallel, the **jobs** table is designed to manage job postings issued by the firm. Key attributes include the job title, department, employment type, required qualifications, and salary range. This table is linked via a one-to-many relationship to the **job_applications** table, enabling each job listing to receive multiple applications. The **job_applications** table holds extensive applicant data, including names, resumes, cover letters, portfolios, LinkedIn URLs, expected salary figures, and relevant work experience, providing a comprehensive foundation for recruitment and HR assessment.

Additionally, the database includes two standalone tables: **projects** and **services**. These are not currently relationally linked to other tables. The **projects** table is used to showcase completed or ongoing work, categorized by type (e.g., *interior*, *construction*), and includes attributes such as project title, location, description, and associated images. The **services** table enumerates the various services offered by the firm, such as engineering design or project management—and includes a unique slug for frontend routing, descriptive content, and a display order to support structured web presentation.

To emphasize the human aspect of the organization, two additional independent tables, **developers** and **team_members**, store information such as bios, profile images, contact details, and professional credentials. The **developers** table primarily highlights the software and website development team, including GitHub and LinkedIn links, while the **team_members** table presents firm personnel roles, areas of specialization, and years of experience.

Collectively, this Entity-Relationship Diagram (ERD) outlines a robust, modular database structure that effectively supports both public-facing features and internal administrative operations of a construction and design firm. It facilitates user engagement, recruitment processes, service management, project presentation, and staff visibility in a cohesive and scalable manner.

SCREENSHOTS OF THE ACTUAL OUTPUT





With our knowledge we guarantee success

Fusce id hendrerit lectus. Morbi vitae tortor sed turpis feugiat congue

Guided by Engr. Noel R. Villanueva, passionate individuals, fueled by talent, the company has grown stronger with each new challenge

LEARN MORE →

FEATURED PROJECTS

View Projects →



Figure 2.2 - Home Page



Figure 2.3 - Home Page

ABOUT US



The Firm

People

Developers

Account

TRIV Design & Construction

SINCE 2025.

Guided by Engr. Noel R. Villanueva, passionate individuals, fueled by talent, the company has grown stronger with each new challenge.

HISTORY

Figure 3.1 - About Us Page - TheFirm

HISTORY

TRIV Design & Construction is a full-service firm specializing in architecture, interior design, and construction management. We are committed to transforming ideas into functional and visually compelling spaces. With a focus on quality craftsmanship and client collaboration, we ensure every project meets the highest standards. Our team combines creative design with technical expertise to deliver projects on time and within budget. At TRIV, we build not just structures—but lasting value and meaningful experiences.

MISSION

To provide exceptional architectural and construction services that fuse innovative design, functionality, and sustainability. We are committed to exceeding client expectations through collaborative processes, high-quality workmanship, and ethical practices.

VISION

To be a leading force in the design and construction industry, recognized for shaping spaces that inspire, endure, and elevate the way people live and work.

Figure 3.2 - About Us Page - The Firm

Our Team

Meet the talented individuals who make TRIV Design & Construction a leader in the industry.

Developers

Account



Engr. Noel R. Villanueva

Founder | Consultant



Arch. Ma. Alyza Linelle L. Villanueva, RMP

Founder | Principal Architect

Figure 3.3 - About Us Page - People



Founder | Principal Architect



**Engr. Jan Alison Lynwhel
L. Villanueva**

Founder | Site Engineer

Figure 3.4 - About Us Page - People



People

Developers

Account

Developers

Meet the talented individuals who created this website.



**Noel Andrae Lara
Villanueva**

Full Stack Developer



Lance Aidrian Benico

Full Stack Developer

Figure 3.5 - About Us Page - Developers



Full Stack Developer



Lewis Leander Gecarane

Full Stack Developer

Figure 3.6 - About Us Page - Developers

SERVICES

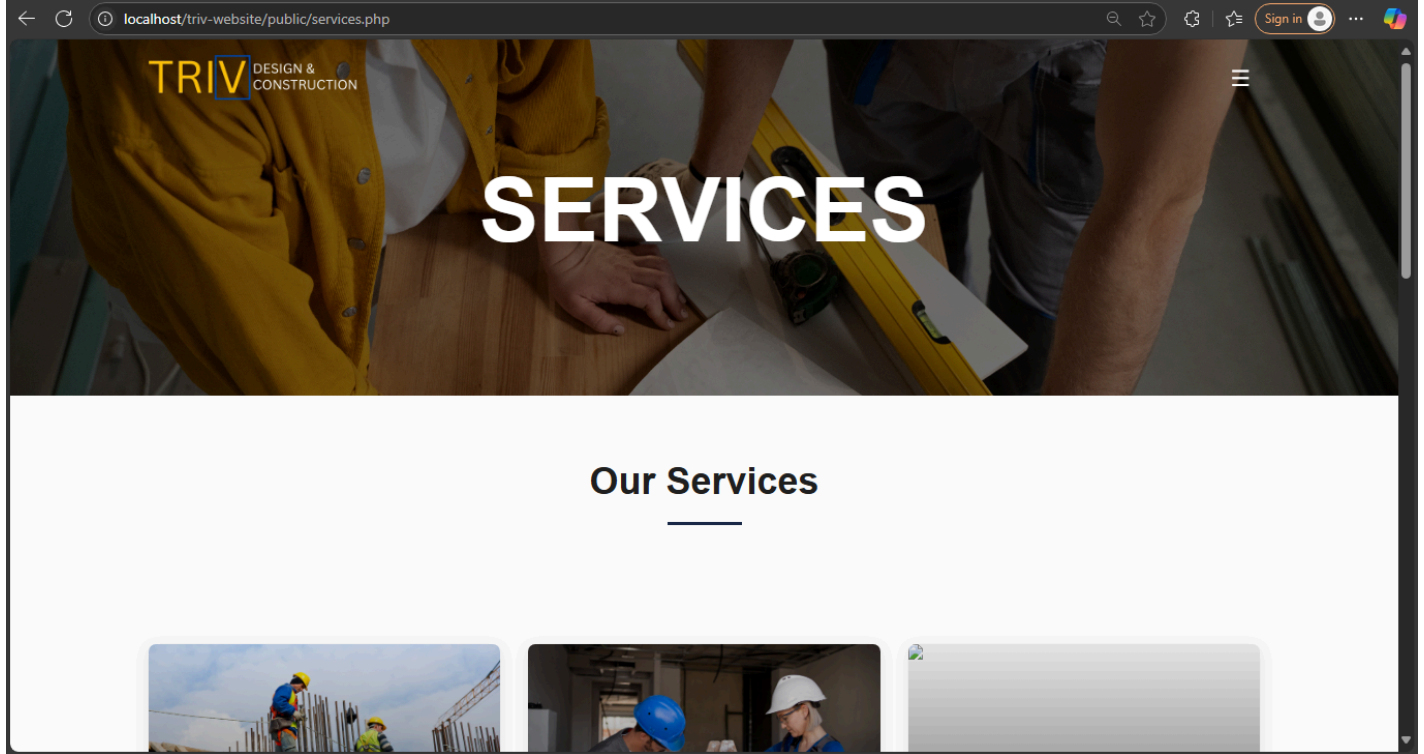


Figure 4.1 - Services Page

ARCHITECTURAL DESIGN

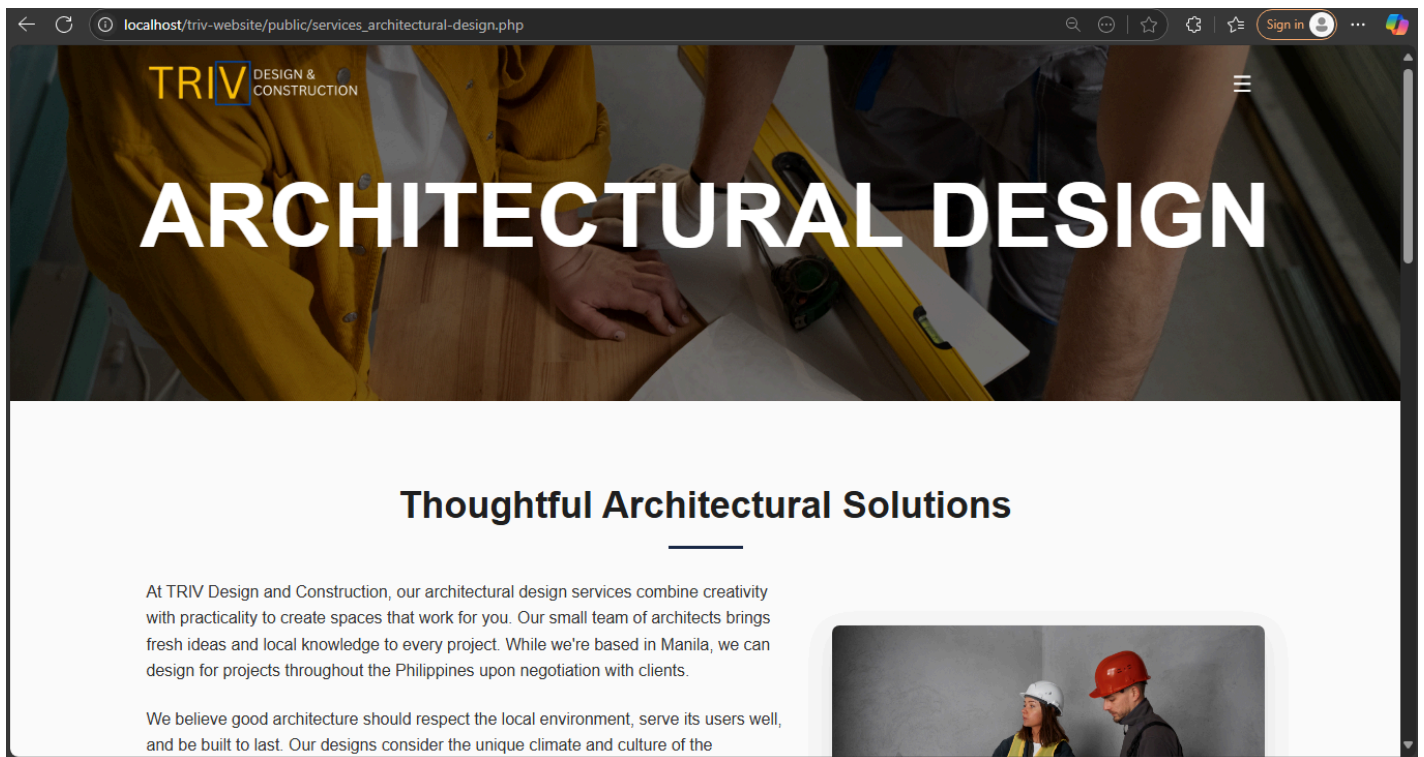


Figure 4.2 - Services Page - Architectural Design

CONSTRUCTION

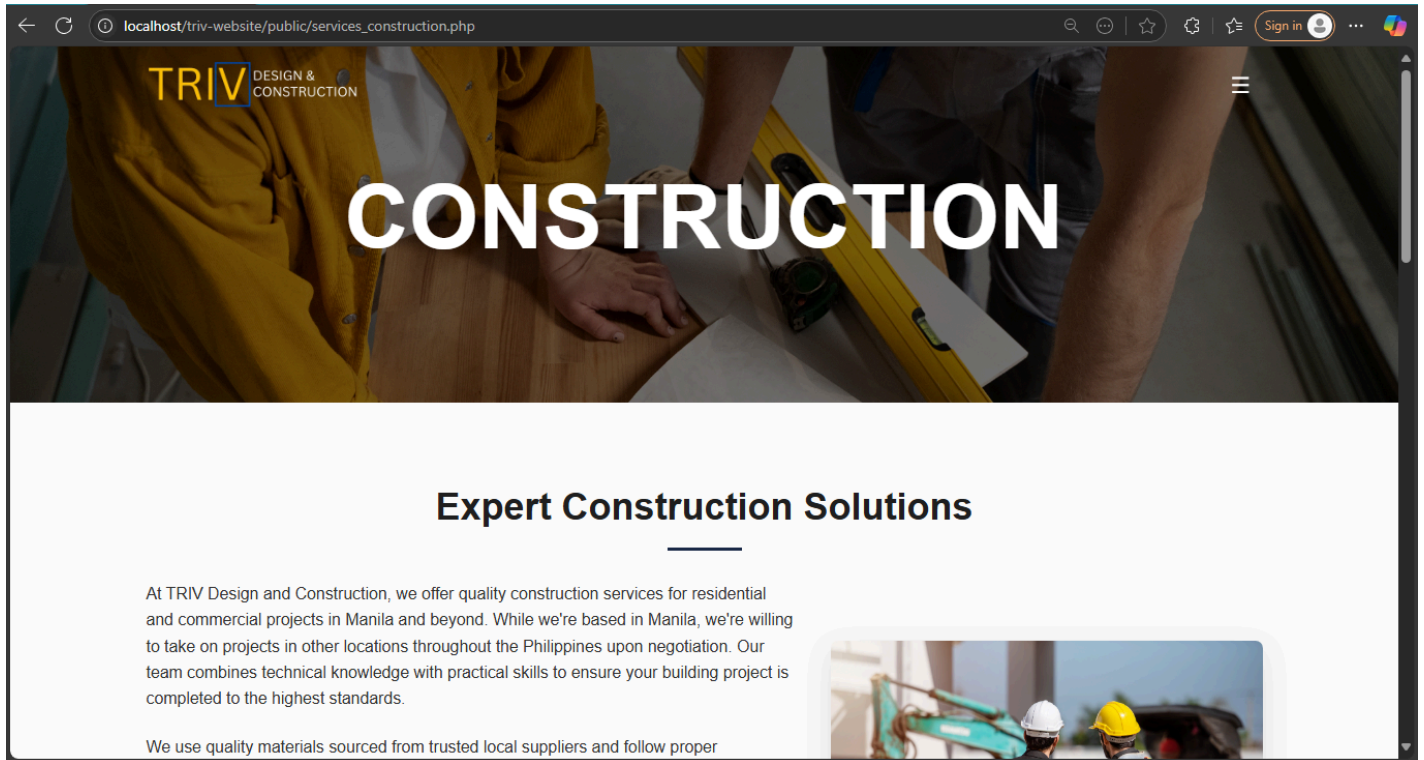


Figure 4.3 - Services Page - Construction

EXTENSION

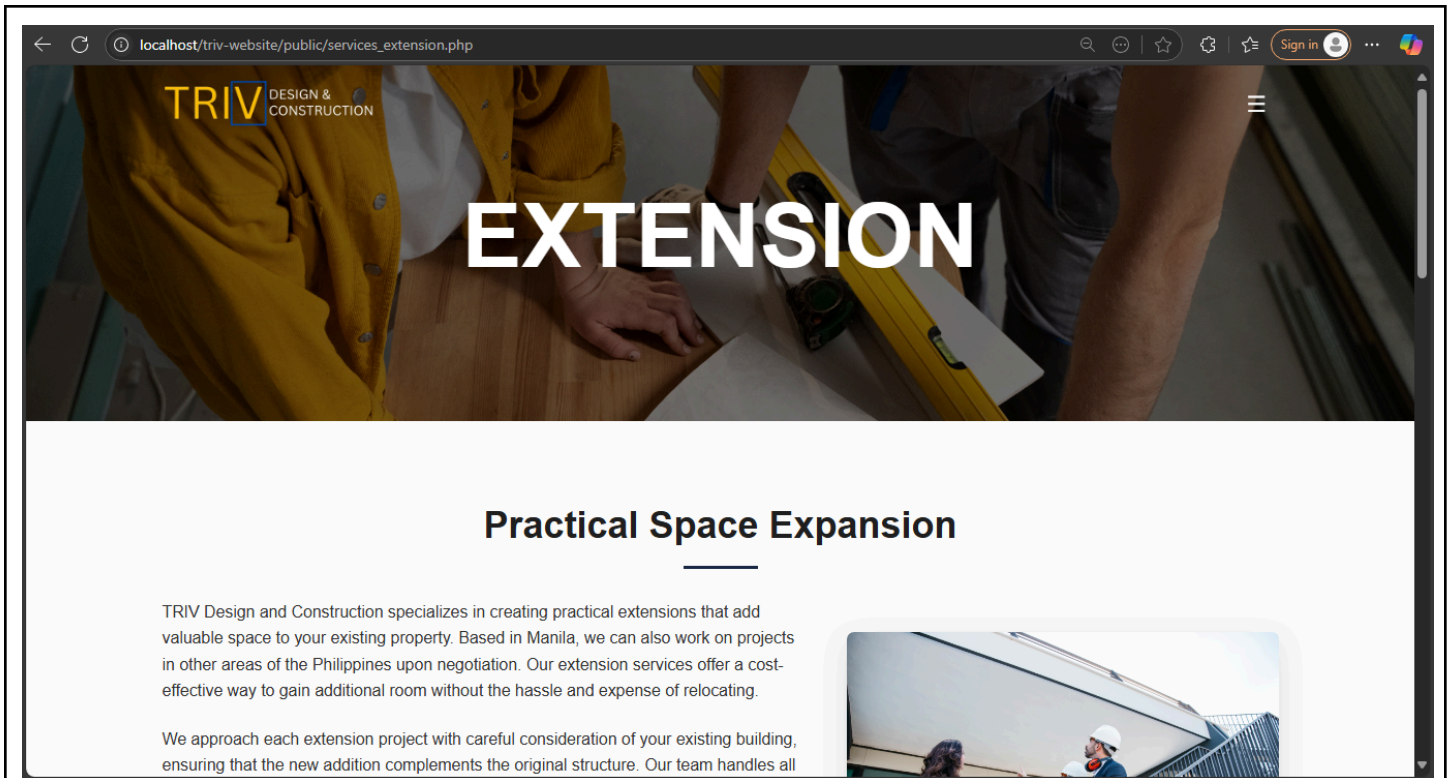


Figure 4.4 - Services Page - Extension

INTERIOR DESIGN

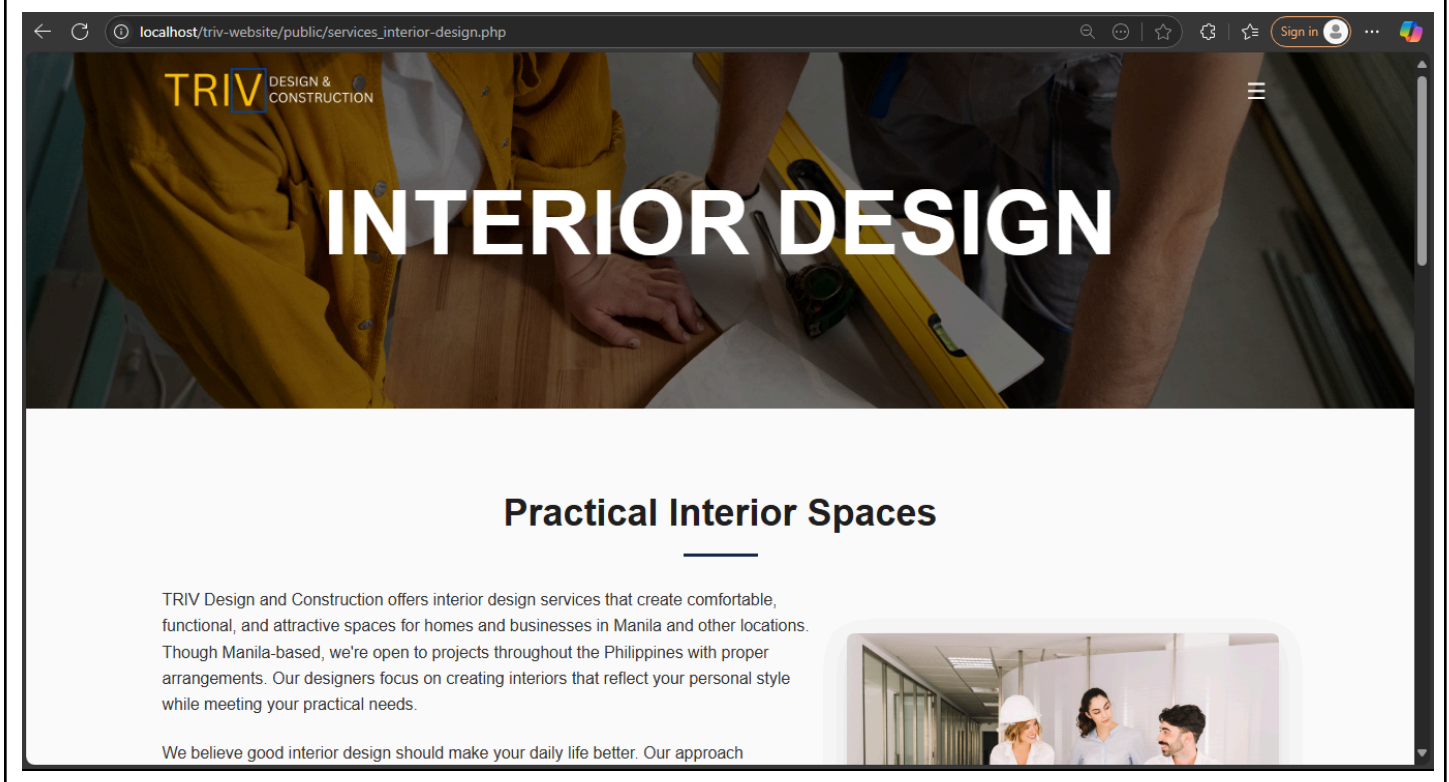


Figure 4.5 - Services Page - Interior Design

RENOVATION

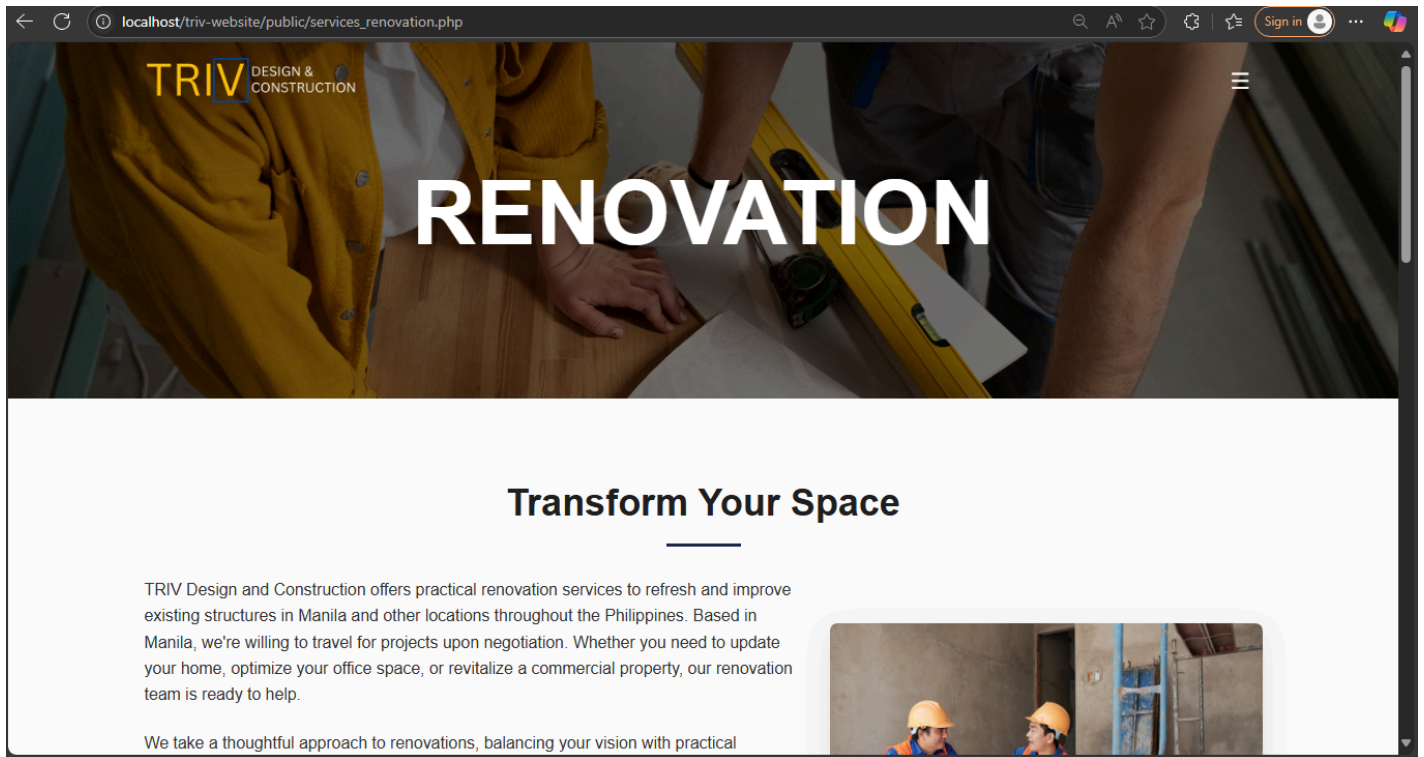


Figure 4.6 - Services Page - Renovation

PROJECTS

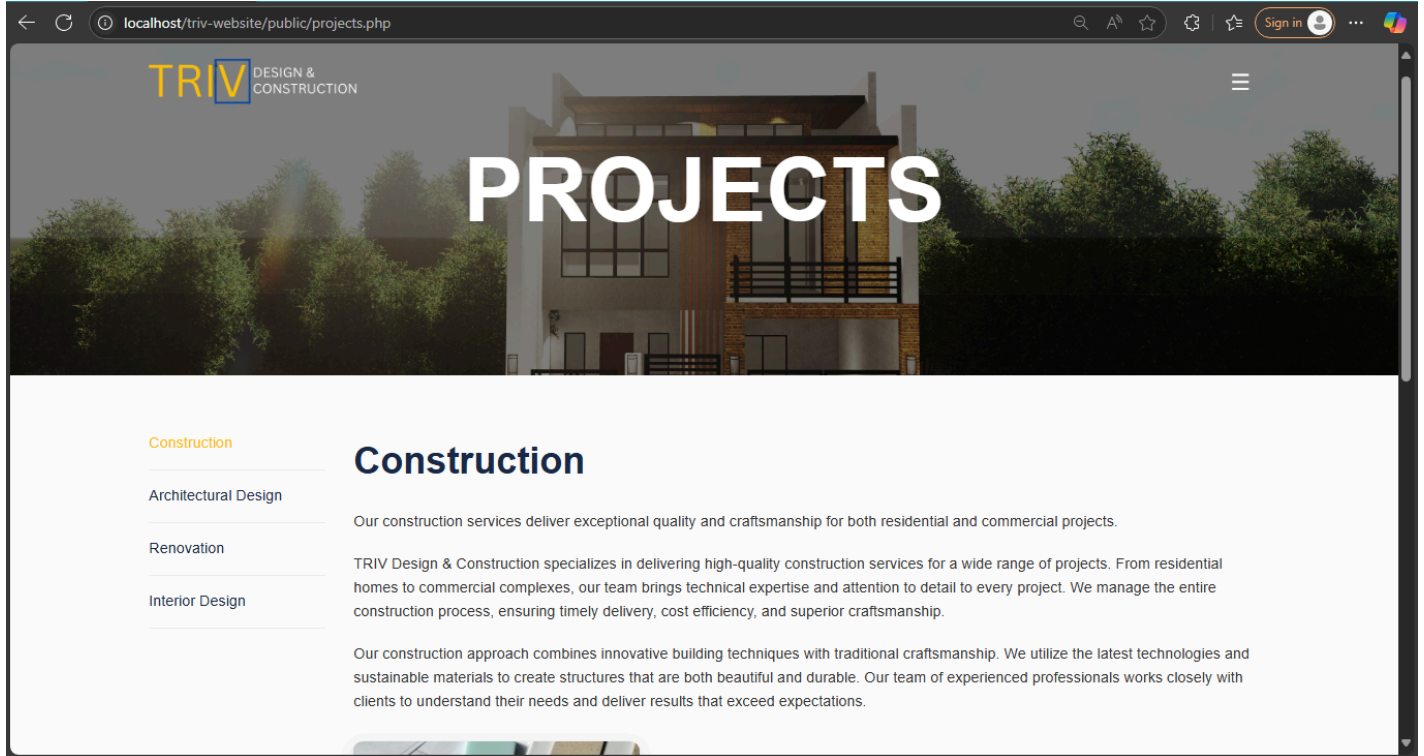


Figure 5.1 - Projects Page

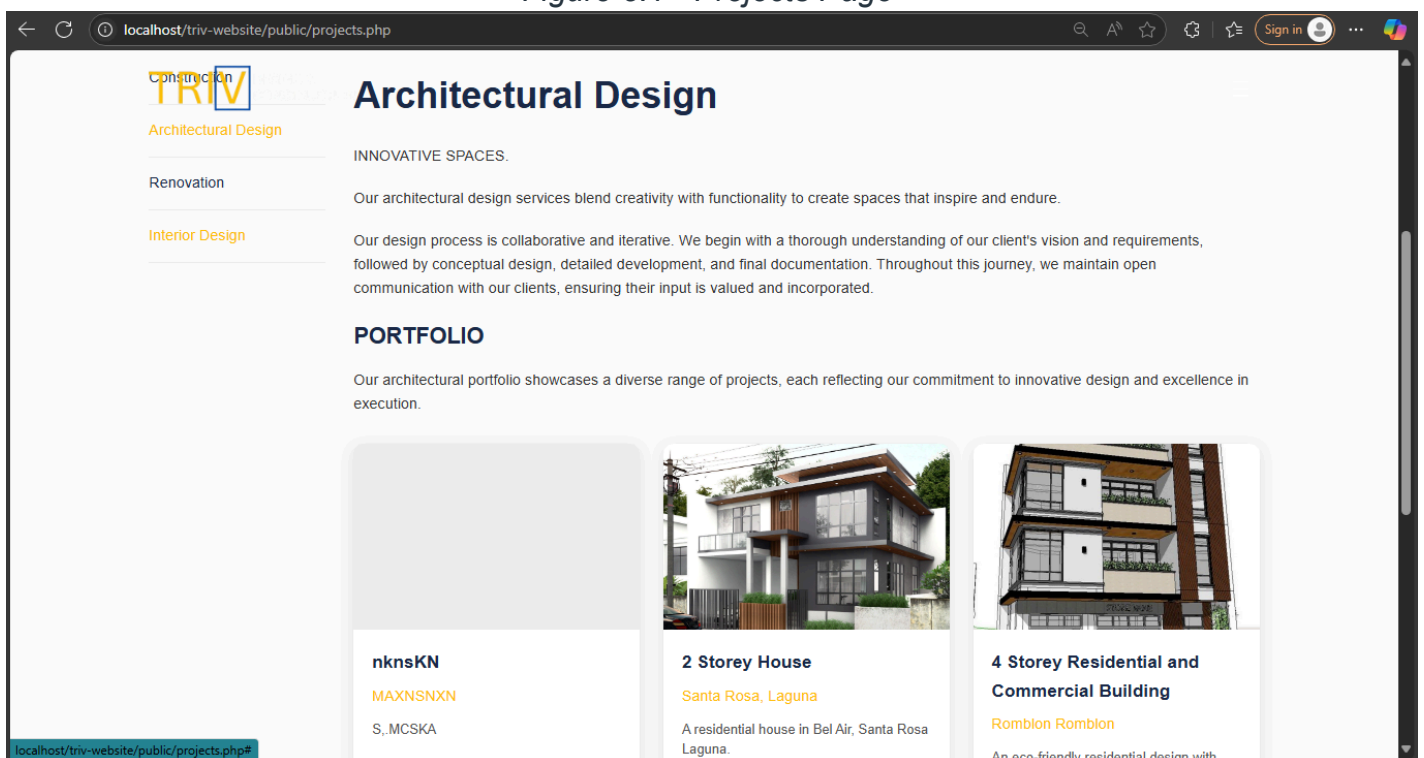


Figure 5.2 - Projects Page

CAREERS

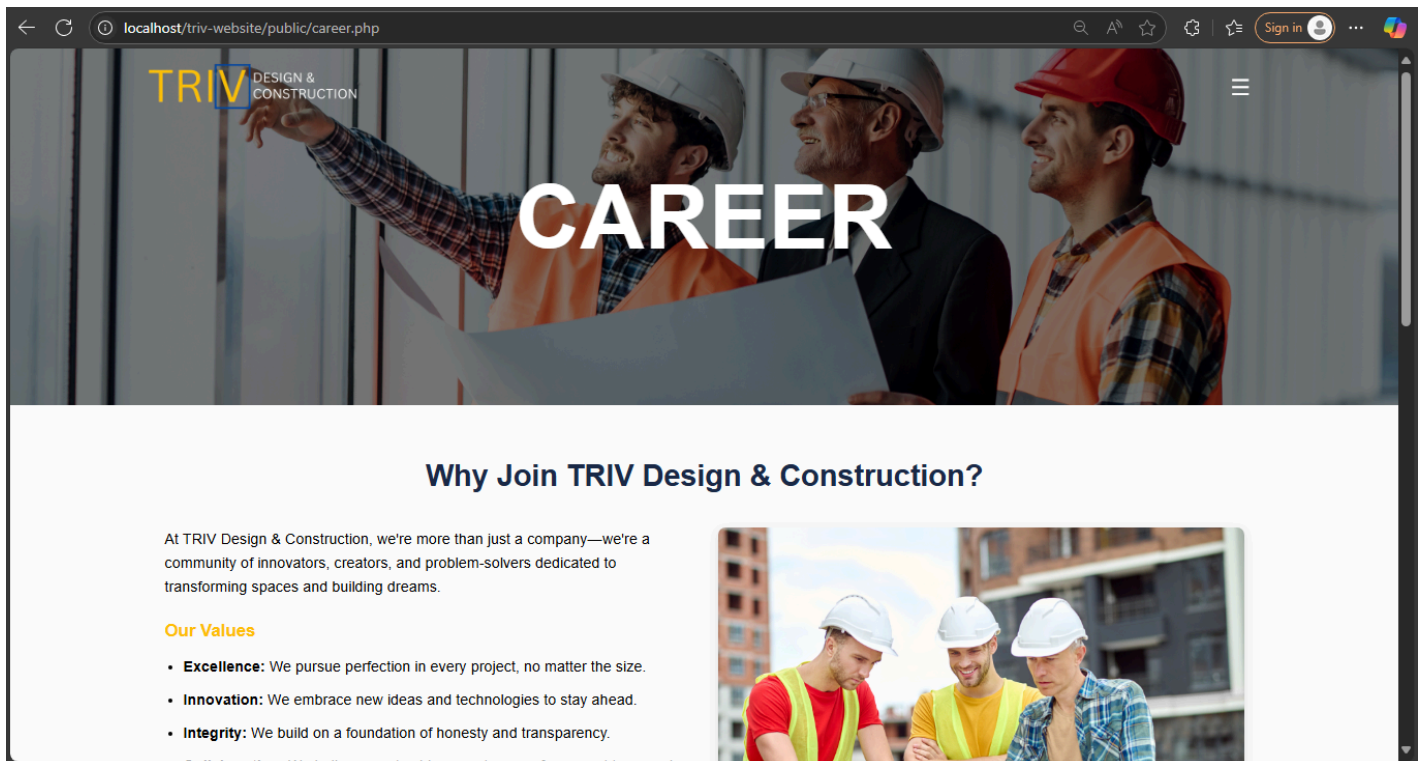


Figure 6.1 - Careers Page

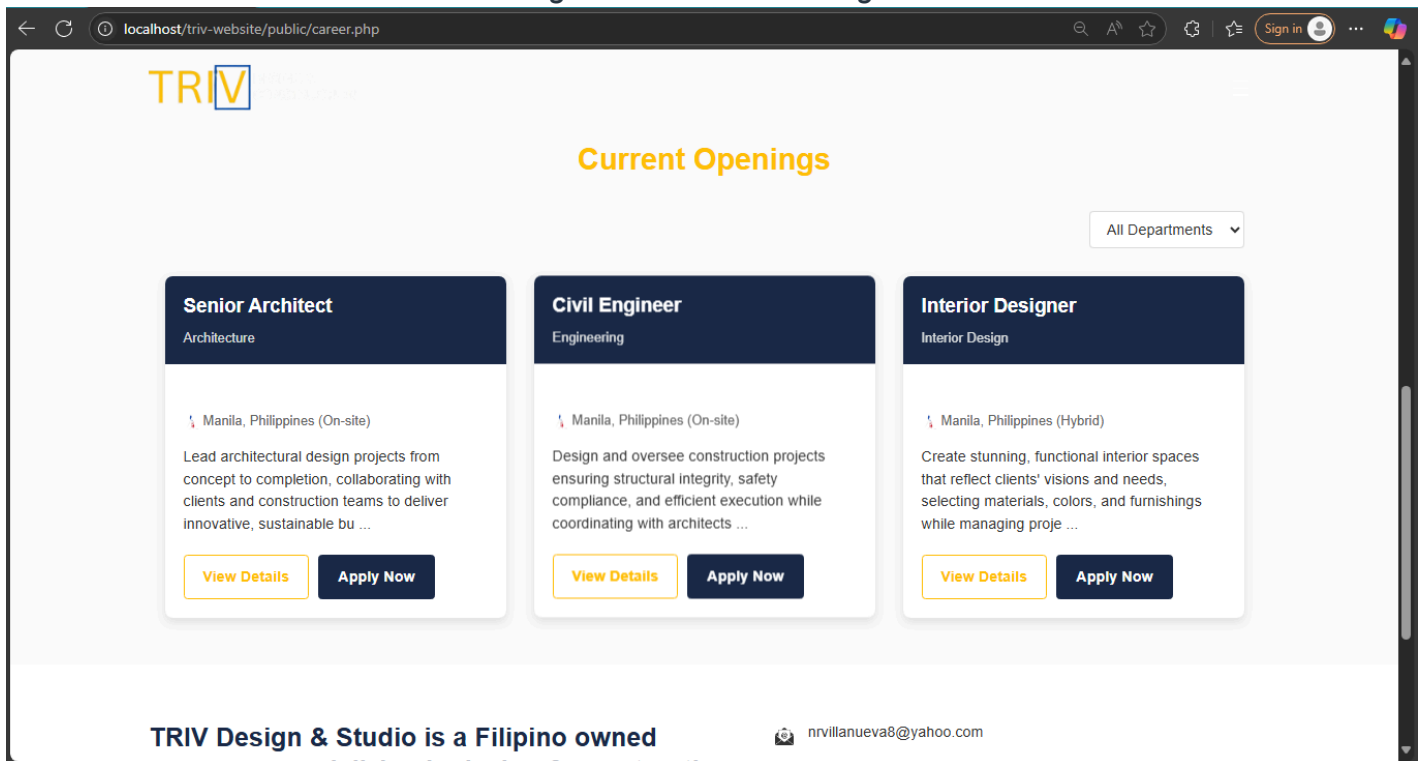


Figure 6.2 - Careers Page

JOB DETAILS

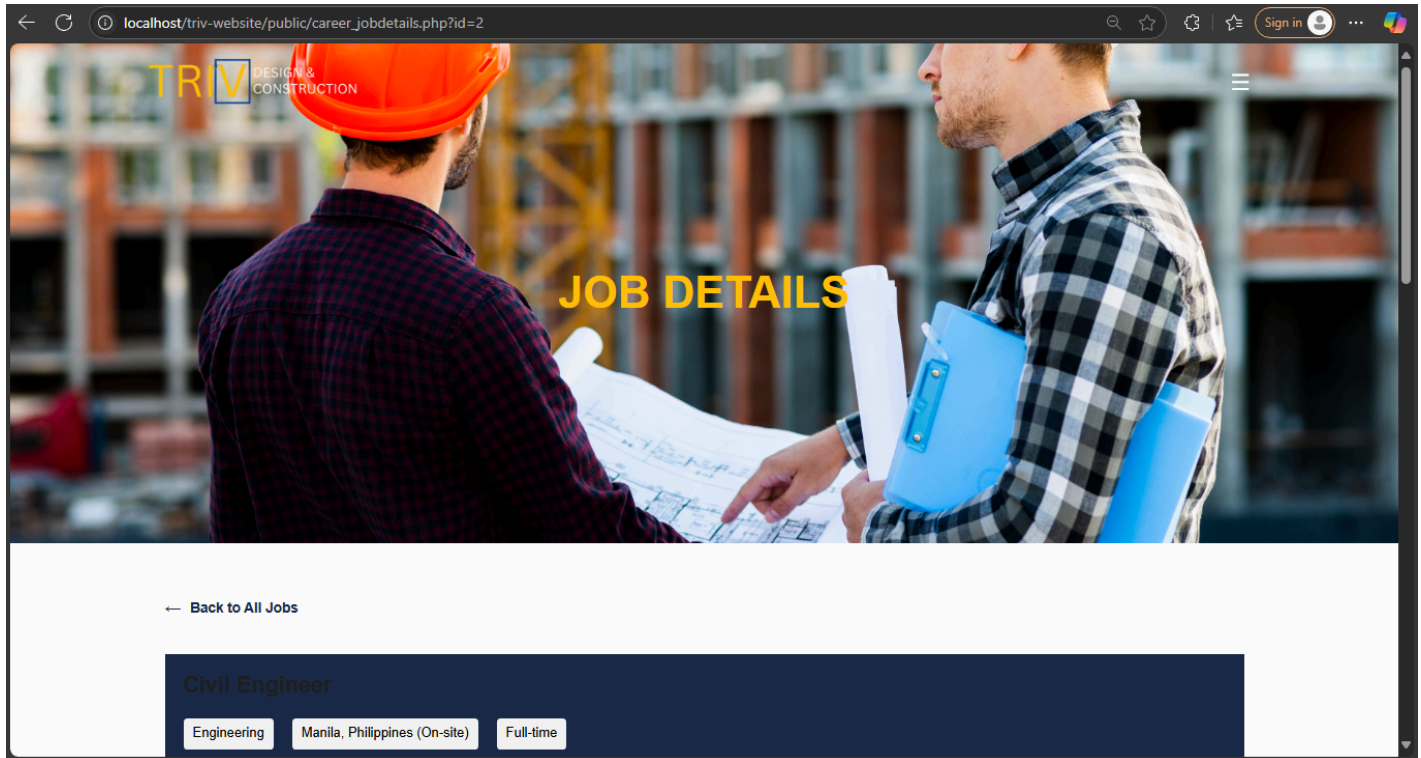


Figure 6.3 - Careers Page - Job Details

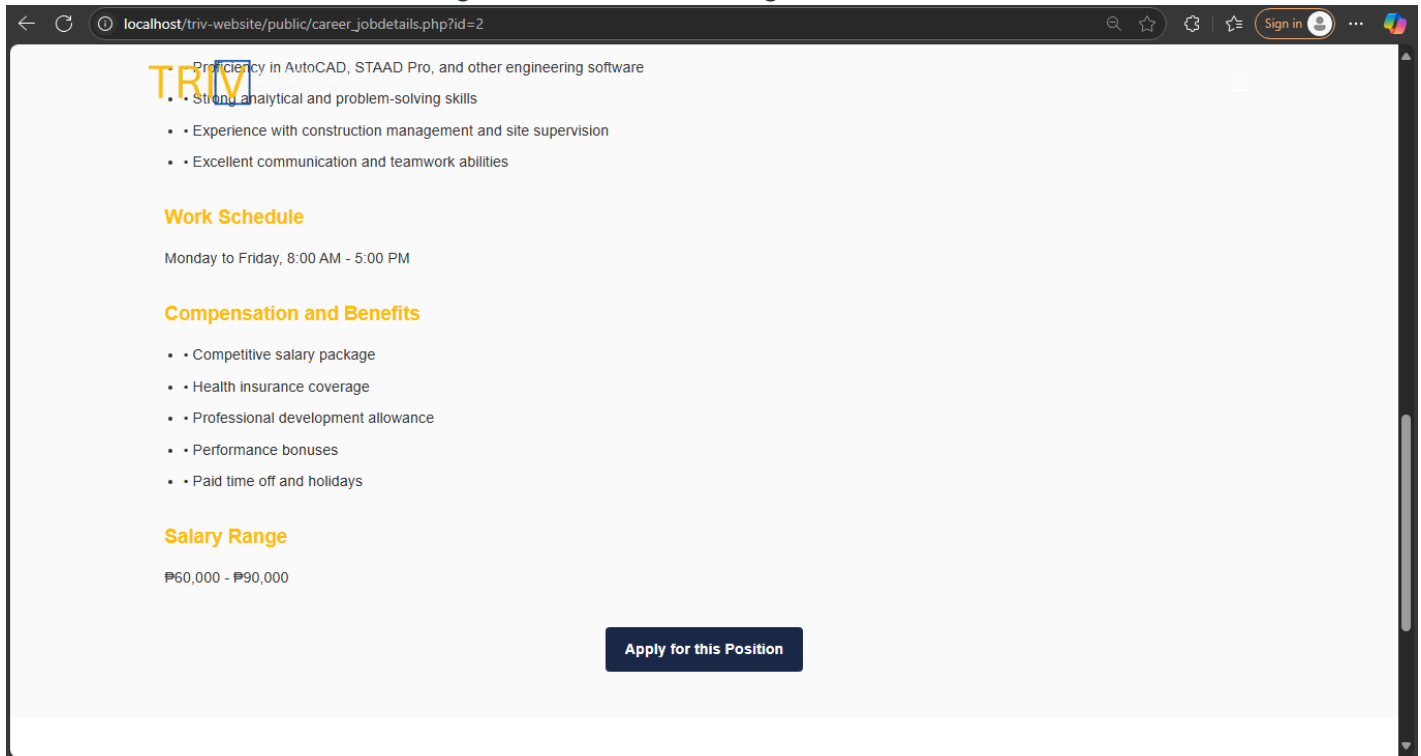


Figure 6.4 - Careers Page - Job Details

APPLY

← Back to All Jobs

Apply for: Position

Please complete the form below to apply for this position. All fields marked with an asterisk (*) are required.

Figure 6.5 - Careers Page - Apply

← Back to All Jobs

Apply for: Position

Please complete the form below to apply for this position. All fields marked with an asterisk (*) are required.

Personal Information

First Name *	Last Name *
Email Address *	Phone Number *
Address *	

Figure 6.6 - Careers Page - Apply

CONTACT US

TRIV DESIGN &
CONSTRUCTION



CONTACT US

Name:

Email:

Mobile:

TRIV Design & Construction



322 National Highway, Masaya Rosario,
Batangas



09087420857

Figure 7.1 - Contact Us Page

Your Message **TRIV** DESIGN & CONSTRUCTION

09087420857

Drop Plan Here

Submit

TRIV Design & Studio is a Filipino owned company specializing in design & construction services.

 nrvillanueva8@yahoo.com

 0908-742-0857

 322 National Highway,

Figure 7.2 - Contact Us Page

FOOTER

TRIV Design & Studio is a Filipino owned company specializing in design & construction services.

 nrvillanueva8@yahoo.com

 0908-742-0857

 322 National Highway,
Masaya, Rosario Batangas

The company brings together a highly skilled technical staff to ensure that excellence in design services standards are achieved to the satisfaction of the clients and owners.

TRIV DESIGN & CONSTRUCTION

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Figure 8.1 - Footer